

Model updating of fluid-structure interaction effects on piping system

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ABSTRACT

The main goal of this paper is to propose a simplified model to predict the dynamic behavior; mainly vibration under effects of fluid. Firstly, an experiment is carefully designed incorporating important parameters of the system and experimental modal analysis is performed for solid only and with fluid. Then, modal analysis using FEM is performed in real scale 3d model of the test rig and the obtained results is approximated to experimental frequency of the system. Then, a simplified model is developed, which appends the results of explicitly computed mass and stiffness of supports, to a solid only FEM. The new model is analyzed and updated till it is considerably close to the results obtained previously. Based on sensitivity analysis, set of model parameters is selected for the model updating process. Response surface method is implemented to find values of model parameters yielding results closest to the experimental. The results can provide a basis for further experimental and numerical dynamic analysis and optimization.

Keywords: Fluid-structure Interaction, Added-mass, pipelines

INTRODUCTION

Piping systems are designed so as to transfer fuel fluids reliably when subjected to vibration most of the time. Although traditional vibration isolation techniques are invariably used in existing systems, dynamic analysis of such piping systems have to be carried out to optimize the performance and prevent failure. To accurately predict the hydrodynamic loadings in the fluid as well as the stress levels and vibrations in the piping, experimental and theoretical researches have shown that the analysis of the fluid and pipe motions must be carried out simultaneously in a coupled way. Such kind of approach is referred to in the literature as fluid-structure interaction(FSI). These coupled analyses involve solving computationally costly equations in addition to prevalent structure only and fluid only FEM. When the fluid inside the pipe is very slow or almost stationary, the fluid can be modeled just as added mass. This paper intends to provide expressions for explicit added mass of fully filled pipe in free and simply supported conditions in terms of density of water.

BACKGROUND

Laithier and Païdoussis described and derived a general formulation of the dynamics of fluid conveying pipes[1]. Inherent and dynamic characteristics of FSI of pipes needs to be analyzed carefully so as to optimize fluidic and structural design parameters of pipe systems[2–4]. Zhang, Tisjelling and Vardy worked on calculating the natural frequencies of the liquid filled pipes using method of characteristics(MOC) method[5]. Many researches have been conducted to implement Finite element method for calculation of FSI as well[6,7]. The lateral vibrations in pipe can be considered as added mass only according to Dubyk and Orynyak[8]. Li, Karney and Liu have provided extensive literature review on fluid structure interaction in their paper[9]. Escaler, Torre and Goggins analyzed mass effects in free-free condition[10], this paper further extends their research for simply supported condition.



Figure 1 DN32 pipe for free-free simulation



Figure 2: DN50 mounted on support test rig

ANALYSIS

The experiment has been carried out in the Lab at the College of Mechanical and Electrical Engineering in the Nanjing University of Aeronautics and Astronautics. Nominal diameter DN32 and DN50(ISO) of length 1 m steel pipes were used in the experiment. Sponge support were used to simulate free-free conditions for both pipes as shown in Figure 1. For supported pipe test a test rig was built as shown in Figure 2. The test rig is designed with a stiff base plate so as to simulate the fixed base.

Test Rig and Experimentation setup:

7 uniformly separated locations were identified and marked on each pipe along the length so as to obtain the first 3 bending mode shapes in the vertical directions. The accelerometer (PIEZOTRONICS PCB 356A16) was magnetically attached on the second point from one end as shown in Figure 3. All 7 points were excited sequentially, each point being excited 3 times individually.

A measuring cylinder was used to fill water into the pipes. The cap was closed after pouring water into the pipe.

An instrumented hammer; Dytran model 5800B4 was used to excite all configurations. The accelerometer was connected to OROS OR363 system.

Table 1 Dimensions of pipe

Pipe	Outer Diameter (m)	Inner Diameter (m)	Thickness (m)	Cap mass (Kg)	Effective length (m)	Liquid Volume (m ³)
DN32	0.0424	0.0359	0.00325	0.09405	1.04	0.001011
DN50	0.0603	0.053	0.00365	0.18905	1.04	0.002205

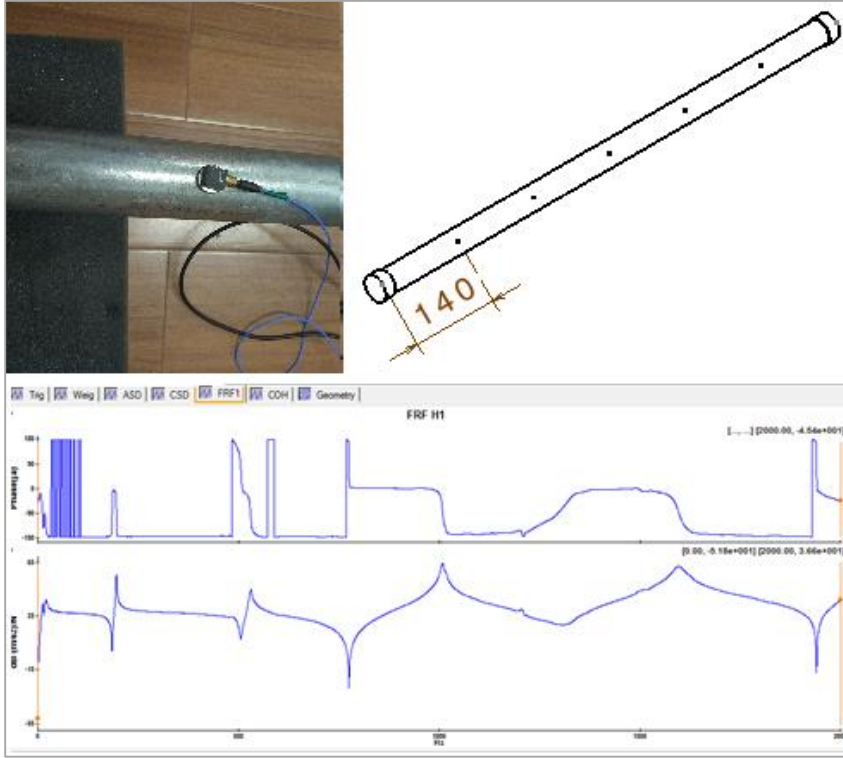


Figure 3: Data acquisition

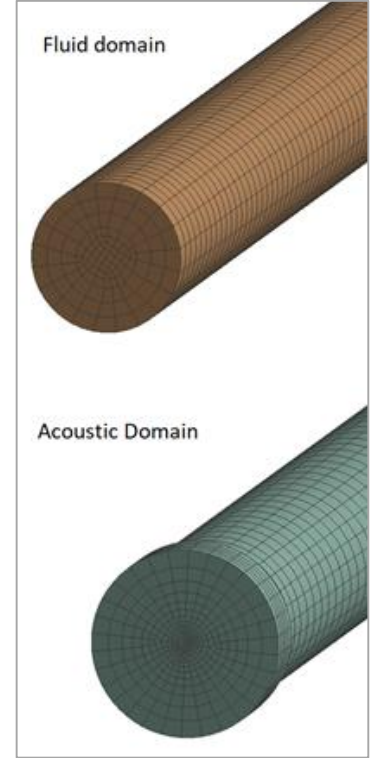


Figure 4 Mesh of acoustic and fluid domain

Numerical Simulation

Acoustic Structural FEA

As Escaler, Torre and Goggins¹⁰ pointed out earlier Navier-Stokes(NS) equations are coupled with structural dynamics equation to solve coupled Acoustic-Structural analysis. The structural deformation and pressure waves generated in the fluid act simultaneously on each other. The unsymmetrical coupled FSI matrix system as mentioned in Eq. 1 can be obtained from combination of discretized structural and lossy wave equations.

$$\left(\omega^2 \begin{bmatrix} \mathbf{M}_s & \mathbf{0} \\ \rho_0 \mathbf{R}^T & \mathbf{M}_f \end{bmatrix} + j\omega \begin{bmatrix} \mathbf{C}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_f \end{bmatrix} + \begin{bmatrix} \mathbf{K}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_f \end{bmatrix} \right) \begin{Bmatrix} \mathbf{u} \\ \mathbf{p} \end{Bmatrix} = \mathbf{0} \dots \dots \dots (1)$$

where $\mathbf{M}_f$, $\mathbf{C}_f$ and $\mathbf{K}_f$ are the acoustic fluid mass, fluid damping and fluid stiffness matrices, respectively, $\mathbf{R}$ is the acoustic fluid boundary matrix, $\mathbf{f}_f$ is the acoustic fluid load vector, $\mathbf{p}$ is the acoustic pressure vector, $\mathbf{u}$ is the fluid element displacement vector and ρ_0 is the mean fluid density, ω is the angular frequency of the pressure oscillation, and $\mathbf{M}_s$, $\mathbf{C}_s$ and $\mathbf{K}_s$ are the structure solid mass, damping and stiffness matrices, respectively.

Model

ANSYS® v. 18 was used to create the model comprising of structure domain for pipe, cap and supports and acoustic fluid domain for the inner liquid filling the pie as shown in Figure 4. The model was built so as to incorporate the additional length and mass due to cap fittings. Effective length and cap mass along with other dimensions are included in Table 1. Table 2 includes the material properties of Steel, Water and Stainless Steel used for the simulation. Higher order solid elements exhibiting quadratic pressure behavior were used for acoustic domain whereas higher order solid elements exhibiting quadratic displacement were used to model structure domain. Full model analyses were performed with unsymmetric method with acoustic elements and direct method without acoustic elements to extract a mode shapes in the range of 10 to 2500 Hz;

the first 3 vertical bending mode shapes as shown in figure 5 always fell inside this range. To simulate the support of pipes three bush elements were used and parameters were tuned to obtain the closest results to experiment.

Then the structure only model was added with distribute mass as shown in figure 6 to simulate the added mass effect of water. The distribute mass was updated and optimized using Genetic Aggregation RSM and Multi-Objective Genetic Algorithm(MOGA) method built into ANSYS.

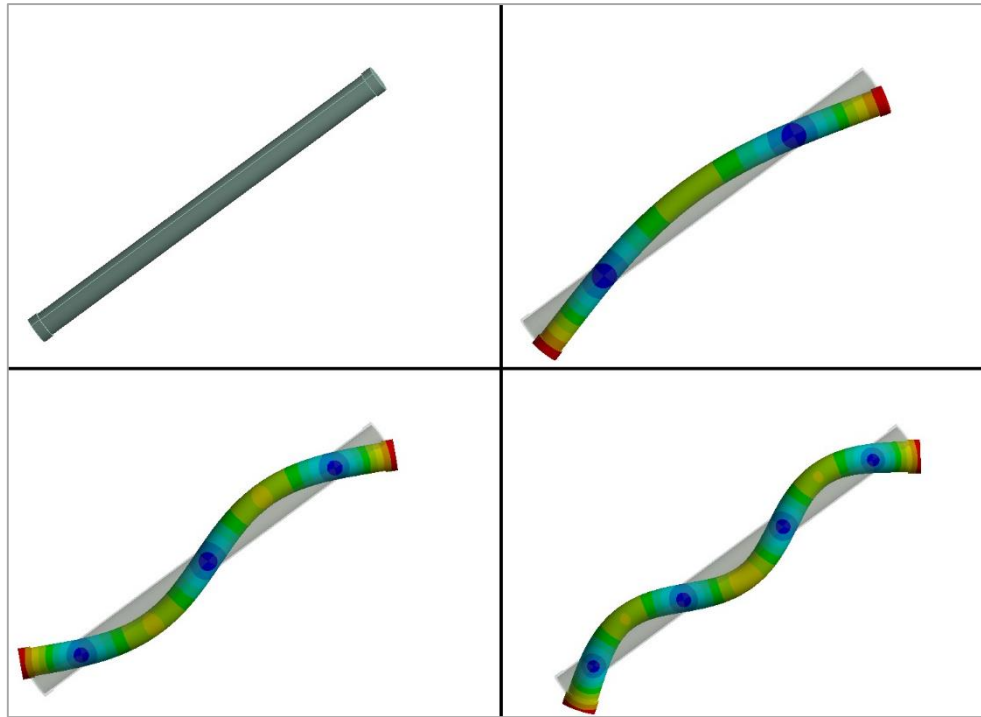


Figure 5: Free mode shapes of the pipe; 1st mode (top right), 2nd mode (bottom left) 3rd mode (bottom right)

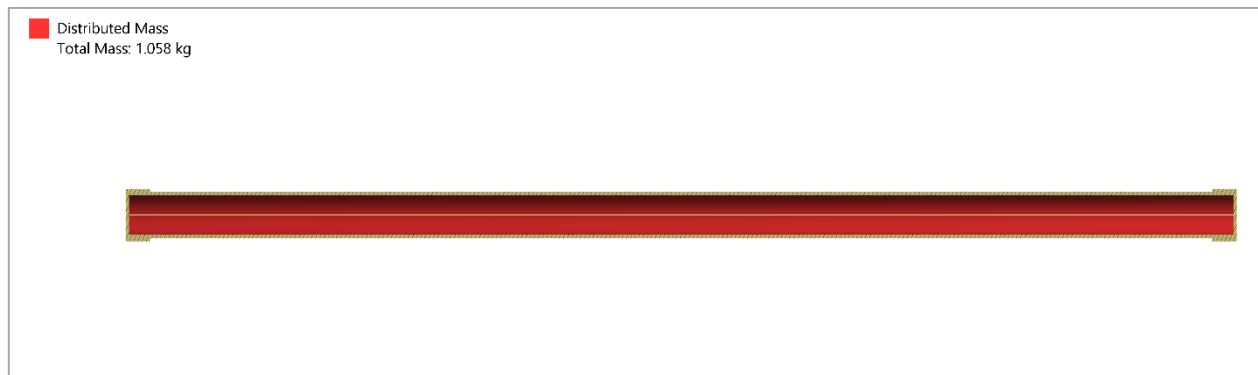


Figure 6: Section view of face selection for distribute mass

Table 2 Material Properties

material	Density(kg/m ³)	Young's Modulus(Pa)	sonic speed (m/s)
steel	7850	2e11	-
water	998	-	1482
Stainless steel	7750	1.93e11	-

Results

The results obtained from the experimental and simulation for free-free condition is presented in the Table 3 and 4.

Table 3 DN32 Free-free condition

Condition	Frequency	f1	f2	f3
without water	EMA(Hz)	224.29	607.83	1160.32
	Sim(Hz)	224.34	613.08	1180
	Dev(%)	0%	-1%	-2%
with water	EMA(Hz)	195.77	528.874	1009.964
	Acoustic Sim(Hz)	202.36	550.82	1061.2
	Dev(%)	-3%	-2%	-5%
	Distributed Mass(Hz)	201.77	548.24	1054.70
	Dev(%)	-3%	-4%	-4%

Table 4 DN50 Free-free condition

Condition	Frequency	f1	f2	f3
without water	EMA(Hz)	311.78	832.013	1543.37
	Sim(Hz)	315.05	849.01	1601.1
	Dev(%)	-1%	-1%	-2%
with water	EMA(Hz)	257.46	692.429	1287.354
	Acoustic Sim(Hz)	269.87	726.15	1368.9
	Dev(%)	-5%	-5%	-6%
	Distributed Mass(Hz)	265.96	713.93	1342.80
	Dev(%)	-3%	-3%	-4%

The results obtained from the experimental and simulation for the supported condition is presented in the Table 5 and 6.

Table 5 DN32 with supports

Condition	Frequency	f1	f2	f3
without water	EMA(Hz)	142.27	608.28	1174.81
	Sim(Hz)	133.73	517.61	1123.8
	Dev(%)	6%	15%	4%
with water	EMA(Hz)	130.93	400.739	1018.853
	Distributed Mass(Hz)	115.63	447.76	972.65
	Dev(%)	12%	-12%	5%

Table 6 DN50 with supports

Condition	Frequency	f1	f2	f3
without water	EMA(Hz)	179.63	854.73	1539.925
	Sim(Hz)	187.70	692.82	1455.5
	Dev(%)	-4%	19%	5%
with water	EMA(Hz)	163.34	716.305	1261.704
	Distributed Mass(Hz)	152.80	564.30	1187.60
	Dev(%)	6%	21%	6%

We can clearly see that in the free-free condition the maximum deviation from experimental and simulated results is 6%. The initial distributed mass was just used as the dead mass i.e. density times volume of the fluid. This result is also included in table 3 and 4. The added mass parameter was then updated against the experimental and acoustic simulation results as shown in figure 7. The updated mass parameter was used to simulate the mass effect on the supported condition. In table 5 and 6 the distributed mass used is the function of updated density(ρ') as shown in Eq 2.

$$\rho' = \frac{DM}{\rho_f} \dots \dots \dots (2)$$

Where DM is the distributed mass used for simulation.

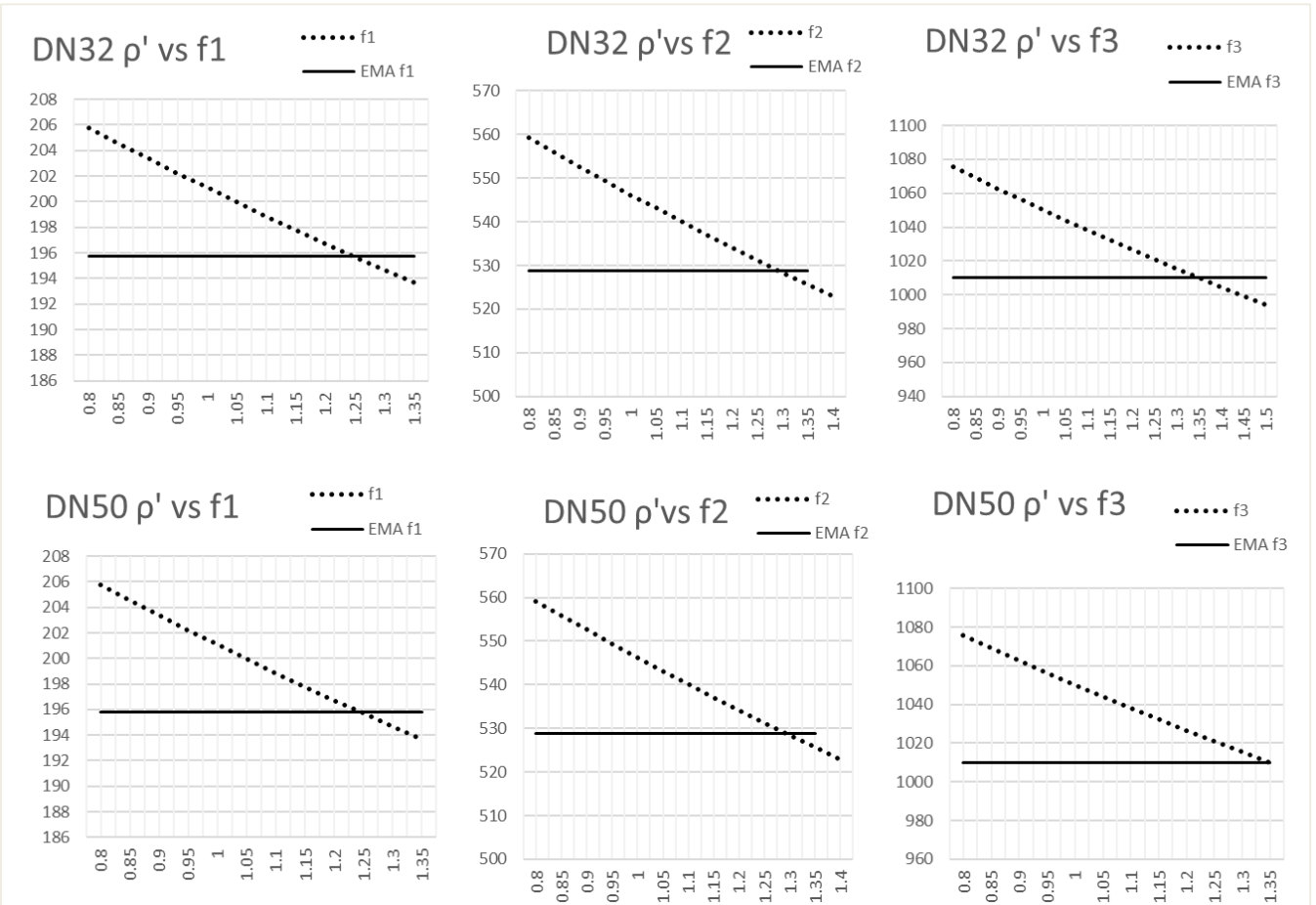


Figure 7 Density coefficient versus frequency

As shown in the figure 7, the 3 bending frequency are almost linearly dependent on the density coefficient within given range. The optimization algorithm figured out 1.4 and 1.2 as optimal coefficient for DN32 and DN50 respectively for fitting the simulation with experimental results.

For supported condition though the deviation between simulation and experiments were up to 15% for cases without fluid and 21% for cases with fluid. The bushing contacts used for simplification of support may have caused the initial aberrance which cumulated into a massive deviation for fluid enclosed condition.

CONCLUSION

In free conditions the additional distributed mass can easily simulate the FSI-mass effect of the still fluid. This method is computationally far less intensive due to low number of elements as a whole and just the use of structural solver. For the supported case the parameters for supports could not be tuned enough to match the experiment without fluid. This may have resulted in overall large deviation for the distribute mass approach in the case with fluid.

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REFERENCES

1. Laithier BE, Paidoussis MP. The equations of motion of initially stressed Timoshenko tubular beams conveying fluid. *J Sound Vib.* 1981;79(2):175-195. doi:10.1016/0022-460X(81)90367-9
2. Hashemi MR, Abedini MJ, Simos TE, Psihoyios G, Tsitouras C. Numerical Modelling of Water Hammer Using Differential Quadrature Method. *AIP Conf Proc.* 2007;936:263-266. doi:10.1063/1.2790125
3. Ibrahim R a. Mechanics of Pipes Conveying Fluids-Part II: Applications and Fluidelastic Problems. *J Press Vessel Technol Asme.* 2011;133(2):1-30. doi:10.1115/1.4001270
4. Yi-min H, Yong-shou L, Bao-hui L, Yan-jiang L, Zhu-feng Y. Natural frequency analysis of fluid conveying pipeline with different boundary conditions. *Nucl Eng Des.* 2010;240(3):461-467. doi:10.1016/j.nucengdes.2009.11.038
5. Zhang L, Tijsseling AS, Vardy AE. FSI analysis of liquid-filled pipes. *J Sound Vib.* 1999;224(1):69-99. doi:10.1006/jsvi.1999.2158
6. Sreejith B, Jayaraj K, Ganesan N, Padmanabhan C, Chellapandi P, Selvaraj P. Finite element analysis of fluid-structure interaction in pipeline systems. *Nucl Eng Des.* 2004;227(3):313-322. doi:10.1016/j.nucengdes.2003.11.005
7. Grant I. Flow Induced Vibrations in Pipes , a Finite Element Approach. 2010:74.
8. Dubyk I, Orynyak I. FLUID-STRUCTURE INTERACTION IN FREE VIBRATION ANALYSIS OF PIPELINES. *Sci J Ternopil Natl Tech Univ* 2016,. 2016;(March).
9. Li S, Karney BW, Liu G. FSI research in pipeline systems - A review of the literature. *J Fluids Struct.* 2015;57:277-297. doi:10.1016/j.jfluidstructs.2015.06.020
10. Escaler X, De La Torre O, Goggins J. Experimental and numerical analysis of directional added mass effects in partially liquid-filled horizontal pipes. *J Fluids Struct.* 2017;69(January):252-264. doi:10.1016/j.jfluidstructs.2017.01.001